PUYA FARD

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Embedded Software Engineer focused on low-level networking and cloud-connected firmware on embedded Linux. Work spans hardware acceleration, kernel modules, and client-to-router and router-to-router data aggregation across MediaTek and Qualcomm chipsets on the Deco mesh router product line. My motto is to keep learning and sharpening my engineering mindset — as my mentor said, an engineering mindset is a lifelong cycle, and I'm eager to keep learning from harder problems in both work and life.

WORK EXPERIENCE

Embedded Software Engineer | TP-Link Systems Inc. — Irvine, CA

Jan 2025 – Present

Work across the QoS, HomeShield, networking, and cloud modules of the Deco mesh router product line, contributing to project lifecycle management for remodel projects, cross-team coordination, and mentoring a new junior hire. Recipient of the TP-Link CEO Award of Excellence (2025) for ownership, technical contribution, and work ethic.

Platform & ownership

- Develop new features, fix bugs, and maintain production firmware for the Deco mesh router fleet across MediaTek (BE25, BE26, BE13000) and Qualcomm (BE65, BE85) chipsets.
- Drive project lifecycle for remodel projects from feature design through firmware delivery. Owner of BE13000v2 remodel project, sold at Costco nationwide.
- Mentor a newly hired junior engineer on embedded Linux debugging, OpenWrt build architecture and workflows, and setting up complex test topologies using MikroTik RouterOS.
- Serve as the firmware point of contact across iOS/Android app, Cloud, QA, System QA, and platform-line teams — unblocking design questions and new-feature work between organizations to keep release schedules on track.

Low-level networking & hardware acceleration

- Work directly with the MediaTek (MT798x) and Qualcomm (IPQ53xx, IPQ95xx) SDKs, including hardware acceleration paths through HNAT on MediaTek and QCA-NSS-ECM and PPE modules on Qualcomm.
- Debug complex issues end-to-end — identifying root causes with tools such as GDB and SLUB, then delivering fixes in a timely manner.
- Architect cloud-connected subsystems reporting device telemetry and subscription state to TP-Link cloud over NATS, including parental-control cloud-lookup table integration.

Features, tooling, and IP

- Lead AI-powered QoS feature integration across firmware components — including application classification, application DPI marking and handling, and traffic offloading to PPE engines — coordinating a multi-patch Gerrit series spanning userspace and kernel.
- Co-authored two patent filings for HomeShield firmware inventions covering subscription-aware kernel enforcement and cloud telemetry reporting.

Graduate Research Assistant | UC Irvine Pervasive Autonomy Lab — Irvine, CA

Mar 2024 – Dec 2024

- Built a human-aware automotive testbed integrating CARLA simulation with a VR environment under Prof. Salma Elmalaki.
- Assembled and tuned the physical and software stack to match real-world driving feel, then ran human-in-the-loop driving experiments to study driver behavior.

EDUCATION

M.S. Computer Engineering, UC Irvine 2023–2024 • **B.S. Computer Engineering, CSU Fresno** 2021–2023

PROJECTS

SmartZoo Monitoring & Control System

Designed an embedded monitoring system for sensitive zoo habitats (stingrays, reptiles, elephants) combining on-device sensors, embedded controllers, and a cloud backend with a software UI for live data and historical telemetry — the same controller-plus-cloud architecture used in modern industrial site control.

Deep Learning Hardware Accelerators

Designed and optimized hardware accelerators for UNet, VGG16, and a llama3 variant. Achieved up to 70.54% latency reduction and 54.11% energy reduction through dataflow design and iterative profiling.

SKILLS

Languages: C, C++, Python, Lua, and Assembly

Embedded Linux: OpenWrt, QSDK, kernel modules, netfilter/iptables, kernel debugging (GDB, SLUB), UCI configuration, Docker-based build environments, WSL2

Networking: TCP/IP, Ethernet, socket programming, QoS (HQoS, HNAT, TC), DPI, NAT, mesh networking (AWN, APSD), Wi-Fi 6/7/8